higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING (CAD COMPONENT)

(Second Paper)

NQF LEVEL 3

(12050663)

14 November 2019 (X-Paper)
09:00–11:00

REQUIREMENTS: **A4 drawing sheets**
Yellow cover folder (BOE 8/15)
Standard hot-rolled structural steel section table (BOE 8/1)

Calculators may be used.

This question paper consists of 4 pages and 3 diagram sheets.

TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only ONE side of each DRAWING SHEET.
5. Draw a 15 mm border on each DRAWING SHEET.
6. Unspecified radii must be R3.
7. Use your discretion where applicable.
8. Insert ALL loose DRAWING SHEETS in the cover folder – do NOT staple.
9. ALL drawing work must conform to the latest SANS 10111 Code of Practice for Engineering Drawing.
10. A balanced layout is very important and candidates will be penalised for poor planning.
11. Work neatly.

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QUESTION 1: REINFORCED CONCRETE DETAILING

1.1 A 460 mm × 460 mm × 230 mm L-shaped concrete column with 8Y20 longitudinal bars and Y10 stirrups at 160 mm centre must be constructed.

Draw, to scale 1:5, a horizontal cross section through the column (top view) to show the constructional details.

Specifications:

Main RS bars (column):	8Y20
Stirrup:	Y10–160
Concrete cover:	50 mm
Shape code:	74

Label the drawing. (10)

1.2 FIGURE 1, DIAGRAM SHEET 1 (attached), shows an isometric view of a pre-assembled reinforcing cage for pad footing and a bending schedule.

Specifications:

- Square foundation: 1 300 × 1 300 × 400 mm
- Concrete cover: 50 mm
- Main bars: 7Y12
- Column: 250 × 250 (square)

Draw, to scale 1:20, a typical layout of a reinforced concrete foundation pad. Do NOT show column reinforcing details. (15)
[25]

QUESTION 2: STRUCTURAL STEEL WORK DETAILING

2.1 Draw, to scale 1:5, a hold-down bolt.

Specifications:

Base height:	Ø30 mm
Projection:	140 mm
Pocket:	200 mm
Bond length with pocket:	300 mm
Bond length without pocket:	500 mm
Bolt size (shank diameter):	M48
Grete:	30 mm
Sizes NOT given:	Use your discretion

(10)

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2.2 FIGURE 2, DIAGRAM SHEET 2 (attached), shows a simple column base with two views of a connection detail for a portal-frame footing. The base consists of the column, two side (gusset) plates, two base angles, two angle cleats and a base angle.

Draw, to scale 1:10, the following views of the connection detail of portal-frame footings:

2.2.1 Front view (7)

2.2.2 Top view (8)

[25]

QUESTION 3: STRUCTURAL STEEL WORK DETAILING

3.1 Refer to FIGURE 3, DIAGRAM SHEET 3 (attached), and draw, to scale 1:5, typical details of each of the following:

3.1.1 Detail A (7)

3.1.2 Detail B (8)

Specifications:

Top and bottom booms: rolled steel angle profiles, 90 × 90 × 8
ALL internal members: rolled steel angle profiles, 70 × 70 × 8

Gusset plate sizes, as well as information NOT given, are left to your discretion. ALL bolt holes are drilled on the centre lines of the angles and ALL bolts are M16. Insert suitable dimensions. Print a suitable title and scale centred beneath the details.

3.2 FIGURE 4, DIAGRAM SHEET 3 (attached), shows a connection of a steel roof truss. The gusset is 150 mm wide and 300 mm long joining members A and B to the top chord. Members A and B are fixed to the gusset plate with M20 bolts.

Specifications:

Top chord: rolled steel angle back-to-back 80 mm × 60 mm × 6 mm
Ties (A and B): 75 mm × 50 mm × 6 mm

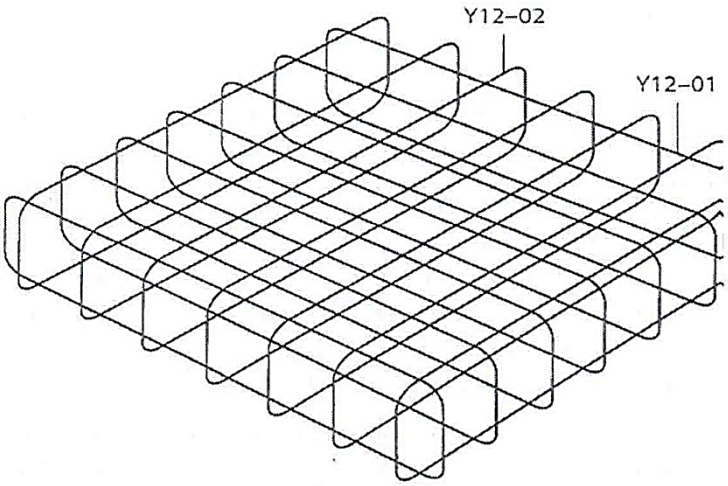
Copy and draw, to scale 1:5, a fully labelled joining detail at 1. (10)

[25]

TOTAL: 75

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DIAGRAM SHEET 1



REINFORCING CAGE FOR PAD FOOTING

BENDING SCHEDULE

Member	Bar mark	Type/ diameter	Number of bars	Cutting length	Bending details
Pad footing	**01**	**Y12**	**7**	**3,20**	
	02	**Y12**	**7**	**3,20**	

FIGURE 1

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DIAGRAM SHEET 2

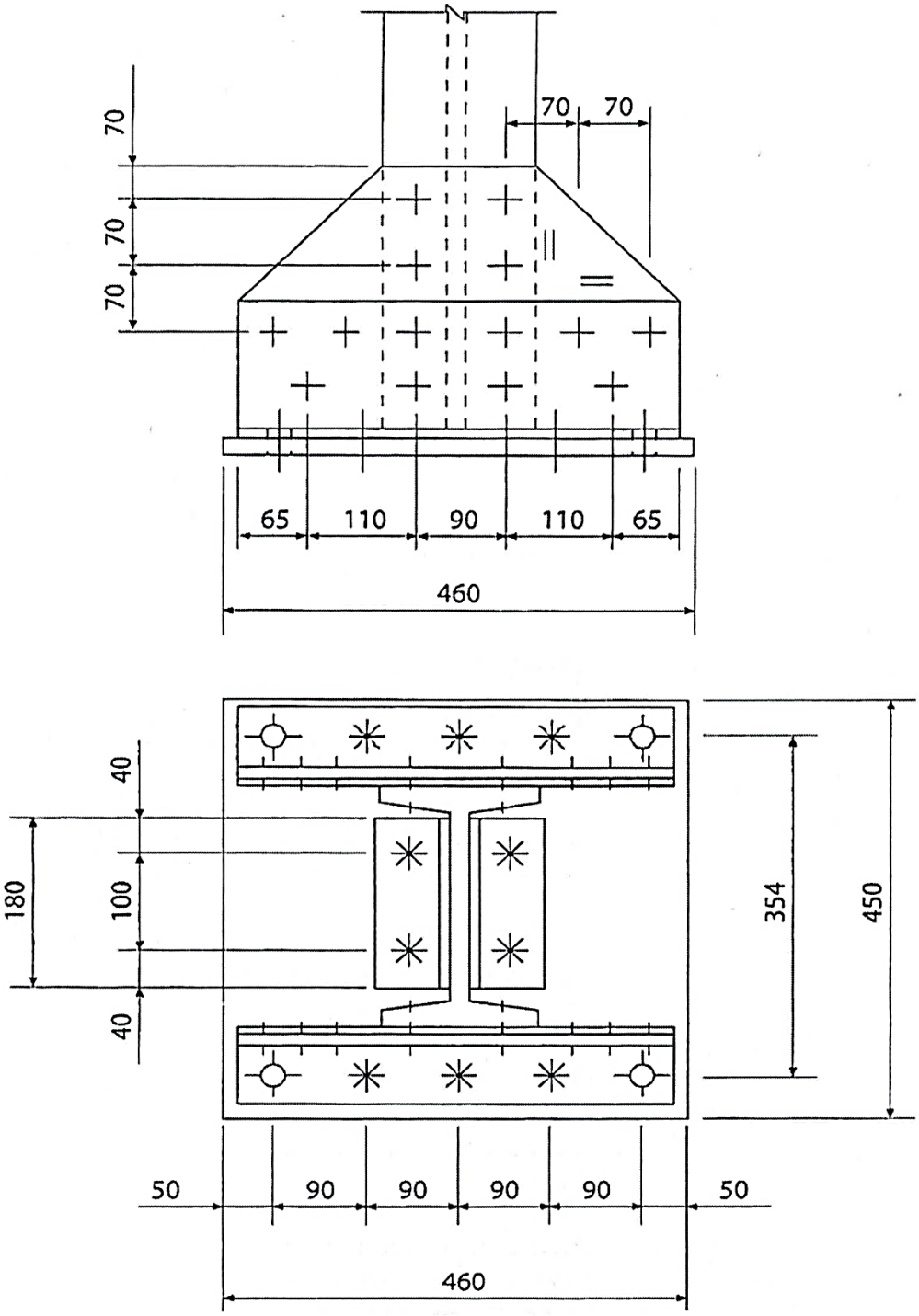


FIGURE 2

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DIAGRAM SHEET 3

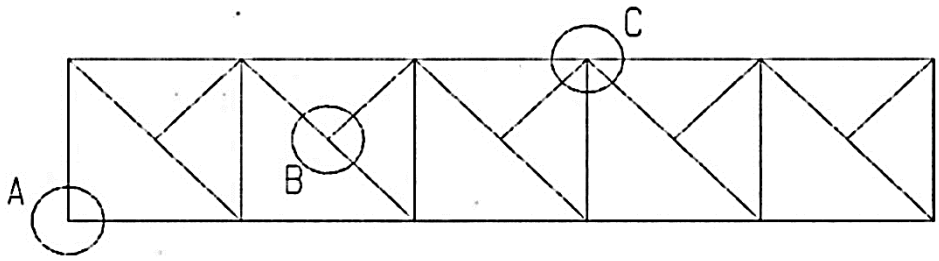


FIGURE 3

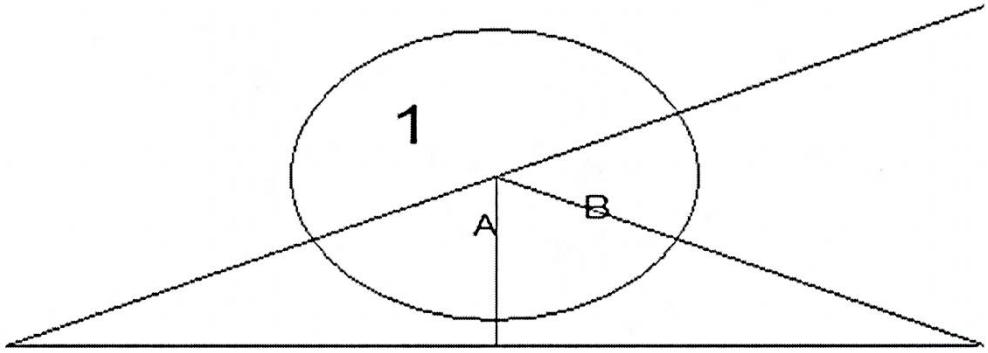


FIGURE 4

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